

Data Type & Data Structure

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September 1, 2026

Outline

Data Type

Data Structure

Lab Report

Basic Data Type

Type	Range	Assign a Value
Numeric	Numbers	<code>x <- 1</code>
Character	Text strings	<code>my_name <- "Rex"</code>
Logical	TRUE or FALSE	<code>Stevenistheteacher <- TRUE</code>

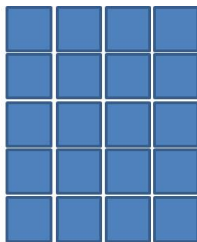
`class()`: To check the data type

Basic Data Structure

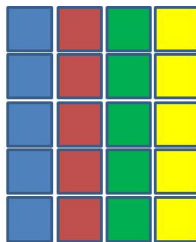
Type	Range
Vector	A list of items that are of the same type
Matrix	A two-dimensional data set with columns and rows
Data.Frame	Data displayed in a format as a table



vector



matrix



data frame

Vectors

`c()` stands for concatenate, which combines elements into a vector

```
score <- c(60, 75, 89, 95, 71, 90)
name <- c("John", "James", "Hank", "Jeff", "
          Ryan", "Ray")
class <- c(1, 1, 1, 2, 2, 2)
```

`rep()`: Replicate elements of vectors and lists

```
rep(x = 1, times = 9)
rep(x = c(1, 2), each = 3)
rep(x = c(1, 2, 3), times = c(2, 3, 5))
```

`seq()`: Tool used to generate regular sequences of numbers

```
seq(from = 1, to = 5, by = 1)
seq(1, 9, by = 2)
seq(0, 1, length.out = 11)
```

Vectors

What will these commands do?

```
score[1]  
score[1:3]  
score[-1]
```



`typeof()` determines the type or storage mode of any object

```
typeof(score)  
typeof(name)
```

You can also use `length()` to know the number of elements in the vector

```
length(score)  
length(name)
```

Exercises: Vectors in R

1. Use the `seq()` function to generate the sequence 9, 18, 27, 36, 45.
2. Use the `rep()` function to generate the sequence 1 1 1 1 1 2 2 2 2 3 3 3 4 4 5
3. Given the following vectors:

```
students <- c("Alice", "Bob", "Chris", "David", "Eve")
```

Write the R commands to:

- 3.1 Extract the first element from the `students` vector.
- 3.2 Extract the first three elements from the `students` vector.
- 3.3 Extract all names from `students` **except** for the second one ("Bob").

Matrices

`matrix(data, nrow, ncol, byrow)` creates a matrix

- ▶ `data`: take a set of values
- ▶ `nrow`: specifies the number of rows
- ▶ `ncol`: specifies the number of columns
- ▶ `byrow`: indicates that the matrix is filled row-wise

```
height <- c(175, 168, 180,  
            154, 185, 162, 170, 187,  
            149)  
matrix(data = height, nrow =  
        3, ncol = 3, byrow = T)  
matrix(data = height, nrow =  
        3, ncol = 3, byrow = F)
```

	Col 1	Col2	Col3
	↓	↓	↓
Row 1 →	1	2	3
Row 2 →	3	4	5
Row 3 →	6	7	8

		Actual Values	
		Positive (1)	Negative (0)
Predicted Values	Positive (1)	TP	FP
	Negative (0)	FN	TN

Matrices

What will these commands do?

```
height.mx[1]  
height.mx[1, 2]  
height.mx[2, -1]
```

Check the structure with

```
dim(height.mx)  
length(height.mx)  
nrow(height.mx)  
ncol(height.mx)
```

	[,1]	[,2]	[,3]
[1,]	175	154	170
[2,]	168	185	187
[3,]	180	162	149

Data Frames

`data.frame()` creates data frames, coupled collections of variables

```
information <- data.frame(  
  Name = name,  
  Class = class,  
  Score = score  
)
```



Data Frames

Name	Team	Number	Age
Defoe	Newcastle	23	33
J. Terry	Chelsea	5	38
Lampard	Chelsea	22	31

Data

Data Frame

What will these commands do?

```
information["Name"]  
information[2]  
information[2, ]  
information[1:3]  
information$Name
```

Check the structure with

```
colnames(information)  
rownames(information)  
length(information)  
nrow(information)  
ncol(information)
```

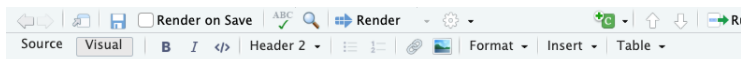
	Name	Class	Score
1	John	1	60
2	James	1	75
3	Hank	1	89
4	Jeff	2	95
5	Ryan	2	71
6	Ray	2	90

Quarto

A framework enables you to weave together content and executable code into a finished document

Mostly contains three parts

- ▶ YAML header
- ▶ Chunks of R code
- ▶ Text mixed with simple text



```
---  
title: "Untitled"  
format: pdf  
editor: visual  
---
```

YAML

Text

Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
{r}  
1 + 1
```

Chunk

Lab Report

Please make sure your R and RStudio are the latest version
Otherwise, Quarto might not work

- ▶ The report file will be in the NTU COOL Assignment section
- ▶ Submit your file in pdf format
- ▶ Deadline: 3/15 (Sun) 23:59
- ▶ Late submission is not accepted

Lab Report

Download the file in NTU COOL

Lab Report 1

Due	Mar 15 by 11:59pm	Points	10	Submitting	a file upload	File Types	pdf
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Questions are in the following qmd file. Please submit your report in the **pdf** file; otherwise, you

No late submission is allowed

[lab_report_1.qmd](#) ↓

Type your student id and name in the author section in YAML

```
---  
title: "Lab Report 1"  
author: "[student_id][your_name]"  
format: pdf  
editor: visual  
---
```

Lab Report

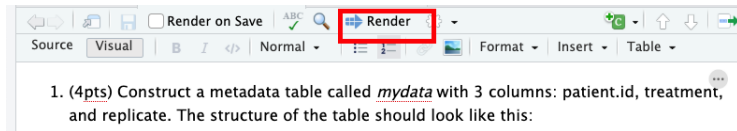
Type in your command in the chunk. You can check your answer via the green arrow

5	C	1
6	A	2

```
{r}
```



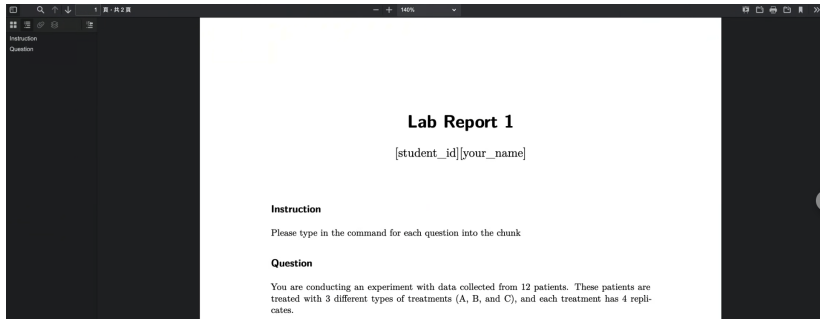
After you've finished all the tasks, click render on the tool list



1. (4pts) Construct a metadata table called *mydata* with 3 columns: patient.id, treatment, and replicate. The structure of the table should look like this:

Lab Report

Download the pdf file and submit it to NTU COOL



*For those who is unable to do it on Quarto, you can also take a screenshot of your code and output, and paste it on a pdf file